Time-varying Mie resonators for real-time manipulation of quantum emitter radiation

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Abstract

Quantum emitters are fundamental to advancing technologies such as secure quantum communication, high-resolution imaging, and quantum computing by enabling on-demand single-photon generation. Precise control over the spectral and radiation characteristics of quantum emitters is crucial for optimizing quantum devices and integrated photonic systems. Nevertheless, traditional designs based on passive materials impose fixed functionality post-fabrication, limiting the ability to tune spontaneous emission dynamically. This study introduces time-varying Mie resonators as a dynamic means to achieve real-time control over the spontaneous emission of a quantum source. A theoretical framework combining the time-Floquet method with multipole expansion is developed to model light-matter interactions in time-modulated environments. Our findings reveal that time modulation enables precise control over multipolar resonant modes, compensating for frequency detuning in the quantum emitter's radiation through an optimal feedback strategy. Moreover, we demonstrate that time modulation can facilitate directional control of far-field emission by selectively exciting specific multipoles and manipulating their mutual coupling. We further show that transverse Kerker conditions enhance radiation directionality and that time modulation can convert nonradiating anapole states into radiating modes via symmetry breaking. Our findings provide a comprehensive framework for real-time manipulation of quantum emitter radiation and open up possibilities for unique applications in integrated quantum physics.

Keywords— quantum emitter, time modulation, scattering, transverse Kerker effect, optical anapoles

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1 Introduction

The emergence of quantum emitters has enabled significant advancements for coupling light and matter, enabling various unique applications such as single-photon light sources [1, 2, 3], quantum computing systems [4, 5, 6], quantum secure communication channels [7, 8], and super-resolution imaging frameworks [9, 10]. Quantum emitters composed of zero-dimensional materials, also referred to as quantum dots, are among the most promising single-photon sources and have been realized through versatile platforms, ranging from atomic emitters or molecules to solid-state emitters [11, 12, 13]. Despite the nature of the quantum emitter, enhanced light-matter interaction is crucial in order to enable directive high-efficiency radiation, which can be attained by exposing quantum dots to minuscule tailored optical structures such as photonic crystals [14, 15] or resonators [16, 17]. In fact, the Purcell effect suggests that the spontaneous emission rate of a single quantum emitter can be desirably modulated by interacting with the enhanced local electromagnetic density of states in resonant optical systems [18, 19, 20, 21]. To give an illustration, by studying a single quantum dot embedded in a photonic crystal waveguide, it is demonstrated that a single exciton in the quantum dot can be recombined by emitting a single photon into a guided mode of the waveguide with a near-unity probability [22]. The powerful electric field enhancement in the gaps of a plasmonic bowtie cavity is depicted to enable the strong coupling regime, thereby facilitating vacuum Rabi splitting even in the individual quantum dot limit [23]. Moreover, in [24], the nonlinear fields of a metallic nanoantenna coupled to a quantum emitter are exploited to open up second harmonic generation, which is inherently prohibited by the symmetry imposed on the problem. Overall, even though plasmonic resonators offer an effective platform for controlling the spontaneous radiation of quantum emitters, their performance is mostly constrained by their intrinsic material loss [25, 26], which necessitates the development of optical structures based on high refractive index dielectric materials.

Dielectric nanostructures have been proven to support optical resonances with high quality factors, making them promising candidates for numerous light-matter interaction applications [27, 28, 29, 30, 31, 32]. The interaction of light with dielectric particles is often described in the context of generalized Lorenz-Mie theory [33, 34], where the electromagnetic fields are expanded in terms of the eigenfunctions of the vector Helmholtz equation, also known as the electromagnetic multipoles [35]. Studying the particles' multipolar response allows for manipulating various resonant modes of magnetic or electric nature, which can be excited inside the resonators to trap and control light at the nanoscale. In fact, depending on the shape, size, and refractive index of the particles, they are able to support different resonances, including dipolar, quadrupolar, and higher-order modes, which their interplay can be engineered to enable applications such as unidirectional light scattering [36, 37], nonradiating optical modes [38, 39], optical forces [40, 41, 42, 43], and asymmetric imaging [44, 45, 46]. To be more specific, transverse Kerker conditions for dipole sources are theoretically derived and experimentally validated by demonstrating that radiation from both electric and magnetic dipole sources can be directed toward their polarization direction by resorting to a dielectric cylindrical scatterer [47]. Nanoantennas with small mode volumes, comprised of high refractive index dielectric gallium phosphide (GaP), are coupled with atomic mono- and bilayers of WSe_2. This configuration significantly enhances the spontaneous emission rate, directs photoluminescence more effectively, and improves optical excitation efficiency. Consequently, the photoluminescence is significantly increased compared to WSe_2 placed on a planar GaP substrate [48]. Additionally, by leveraging the topology of light and matter and manipulating their interaction, nonradiating hybrid anapole states can be excited within a dielectric meta-atom to suppress far-field radiation. It is also demonstrated that by controlling the orientation and inversion symmetry of the particle, it is possible to switch from a nonradiating state to a super-scattering state, where the scattering cross section exceeds the single-channel limit [49]. It is worth noting that these applications are enabled through precise engineering of the particles' inherent attributes, such as their geometrical and material properties, which are fixed after fabrication. Therefore, a practical tuning approach is required to design multifunctional devices with real-time controllable responses.

Among various mechanisms, such as nonlinear effects [50, 51], mechanical reconfiguration [52, 53], and phase-change materials [46, 54, 55], materials with time-varying properties stand out as promising candidates for the design and implementation of active photonic devices with dynamic responses, as they offer a new degree of freedom to the design paradigm [56, 57, 58, 59, 60]. Time modulation can be introduced to photonic structures by varying the permittivity as a function of time through different implementations, such as acousto-optic [61, 62], magneto-optic [63], and free carrier density modulation [64, 65]. In fact, infinite

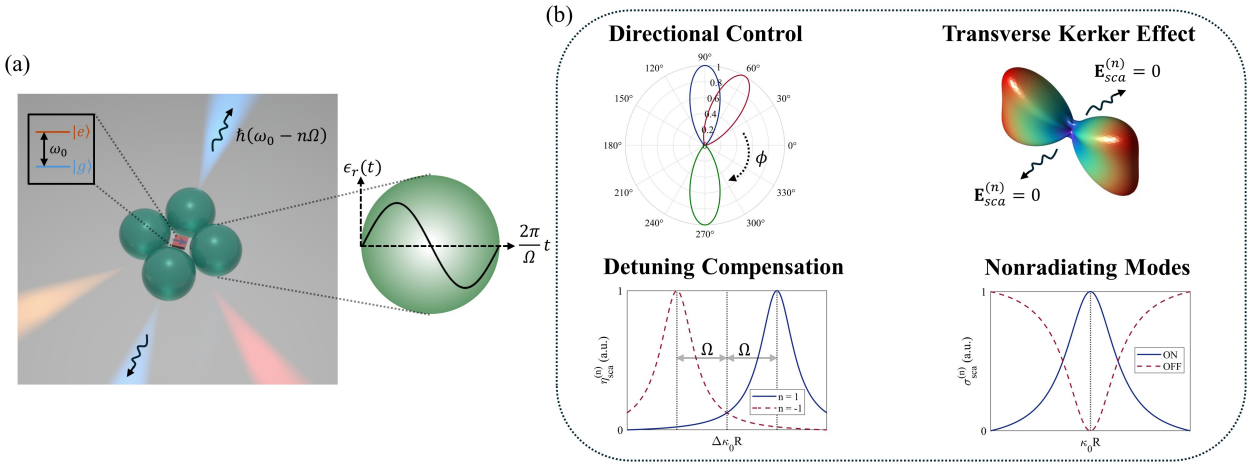


Figure 1: (a) Schematic representation of a quantum emitter positioned at the center of an ensemble of time-modulated resonators. (b) Conceptual illustration of potential applications enabled by time-varying Mie resonators, including directional control, the transverse Kerker effect, frequency detuning compensation, and nonradiating anapole and scattering states.

optical media composed of time-modulated materials, such as time-modulated metasurfaces and temporal photonic crystals [66, 67, 68], give rise to various advanced space-time scattering phenomena and can be envisioned for numerous promising applications, including magnetless nonreciprocal components [69, 70], beam steering [71, 72], active pulse shaping [73, 74], and secure communication [75, 76, 77], to name a few. On the other hand, the interaction between light and individual dielectric resonators with time-varying properties, also known as dynamic meta-atoms [78, 79, 80], is of particular interest, as it not only opens up avenues toward unique applications such as directional control of light [81, 82], parametric resonances and amplification [83], and spectrally reconfigurable quantum emitters [84], but also paves the way for understanding the infrastructural physics underlying more intricate devices that incorporate these elements. As a matter of fact, the interaction between a plane wave and an isolated dielectric spherical particle can be manipulated through time modulation. It has been demonstrated that, with a careful selection of the scatterer's permittivity modulation waveform, both forward and backward light scattering can be achieved by adhering to the appropriate Kerker conditions [81]. The extended boundary condition method, combined with the time-Floquet approach, is implemented to resolve light scattering from a periodically time-modulated object of arbitrary shape. It has been shown that matching the frequency detuning of the incident wave from a particle resonance to an integer multiple of the modulation frequency results in strong inelastic scattering, which is associated with energy exchange between the particle and the electromagnetic field [85]. Furthermore, when the transition frequency of a two-level system is modulated faster than its optical lifetime, the system can be treated as a single-photon source. The spectral content of the emitted photons can be dynamically manipulated through the precise optimization of the modulation waveform [84]. Therefore, we propose and investigate Mie resonators with time-varying attributes to obtain concurrent control over the spectral and radiation characteristics of a quantum emitter, which is modeled through its transition dipole moment. We utilize the time-Floquet method, combined with multipole expansion to develop a theoretical platform in order to model the light-matter interaction between the quantum emitter and resonators. The time-varying Mie coefficients are derived for each individual resonator and the full vector coupling between different particles is outlined by resorting to the multipole translation coefficients. The initial static configuration consists of an ensemble of four spherical dielectric resonators, designed to enhance the spontaneous emission rate of the quantum emitter by resorting to the Purcell effect. It is important to emphasize that this work is primarily a fundamental and conceptual study, focusing on uniform permittivity variations in dielectric spherical resonators to explore the underlying physics of time-modulated light-matter interactions. While spherical resonators provide an analytically tractable framework, our approach can be extended to more practical implementations, such as cylindrical resonators composed of bulk transition metal

dichalcogenides (TMDs) [86, 87], which offer a feasible platform for the experimental realization of time-modulated meta-atoms for quantum emitter control. We demonstrate that, despite being a linear operation, periodic time modulation generates higher-order frequency harmonics or time-Floquet modes, which are up- or down-converted by the modulation frequency, a phenomenon also referred to as the frequency-mixing property of time modulation. We highlight the dynamics of the dipolar Mie coefficients associated with different harmonics in a single time-modulated resonator and illustrate that these coefficients tend to acquire a phase proportional to the harmonic order and modulation phase delay, while maintaining a constant amplitude. An optimal feedback strategy is introduced to compensate for the radiation frequency detuning, which originates from the stochastic nature of the quantum source and its surrounding environment. It is shown that careful selection of the modulation functions can successfully recover the radiation at the desired frequency. Moreover, it is demonstrated that the interaction of the multipolar resonant modes within the resonators can be precisely manipulated not only to steer the radiation toward the desired angles but also to satisfy transverse Kerker conditions and align the far-field pattern with the polarization direction of the quantum emitter. Finally, we exploit time modulation to control the symmetry of the ensemble and demonstrate that it can be utilized as a dynamic switch to excite nonradiating anapole and scattering states and convert them into one another. Our results suggest that time modulation can be successfully applied to manipulate the radiation of quantum emitters and open up crucial applications in integrated quantum optics and photonics. The rest of this manuscript is organized as follows:
In Section 2, we delve into the theoretical platform utilized to obtain the results. Section 3 discusses the design procedure of the static ensemble of resonators. The results for the dynamic control of the quantum emitter's interaction with the resonators through time modulation are presented in Section 4. Finally, the conclusions are drawn in Section 5.

2 Theoretical platform

This section elaborates on the theoretical approach used to model the interaction of a quantum emitter within a cluster of time-varying particles supporting multiple Mie resonances. Figure 2a illustrates the configuration of the problem, which can be generalized to a random distribution of M time-modulated resonators. Initially, we address the light-matter interaction with an isolated time-varying resonator using multipole expansion and the time-Floquet method. Next, we consider the full vector coupling of the particles in the cluster through basis transformation of the electromagnetic fields, utilizing multipole translation coefficients. Finally, the spontaneous emission and far-field scattering characteristics of the system are derived from the obtained electromagnetic fields (comments on the convergence and numerical accuracy of the proposed theoretical platform are available in the Supplemental Material [88]).

2.1 Light-matter interaction with an isolated time-varying Mie resonator

Considering an isolated spherical particle with radius R and relative permittivity $\epsilon_r(t)$, changing periodically with a period T, the electric field can be decomposed into incident, scattered, and internal fields as follows:

$$\mathbf{E}\left(\mathbf{r},t\right)=\sum_{\iota}\mathbf{E}_{\iota}\left(\mathbf{r},t\right). \qquad \iota\in\{inc,sca,int\} \tag{1}$$

The time-dependent permittivity is assumed to be modulated by an external source, which does not directly alter the governing wave equation or interfere with the electromagnetic fields of the resonators, as long as the modulation and field frequencies are distinct. This assumption allows the system to be accurately modeled using a periodic permittivity variation without requiring explicit treatment of the modulation source.
Due to the periodic nature of the problem, and following the time-Floquet approach [85, 89], each electric field contribution can be expanded into a Fourier series as shown below:

$$\mathbf{E}_{\iota}(\mathbf{r},t)=\Re\left\{\sum_{n=-\infty}^{+\infty}\mathbf{E}_{\iota}^{(n)}(\mathbf{r})e^{-i(\omega-n\Omega)t}\right\}, \tag{2}$$

where ω and Ω are the excitation and modulation angular frequencies, respectively. Since the electromagnetic multipoles span a complete space for the solutions of the vector wave equation, they can be utilized to describe

the Fourier coefficients of the electric field [35]. Specifically, the incident and internal field coefficients for each harmonic can be expanded in terms of the regular multipoles as follows:

$$\mathbf{E}_{inc}^{(n)}(\mathbf{r}) = \sum_{\alpha\in\{l,m\}} \left[a_{inc}^{(n)}(l,m)\mathbf{M}^{(r)}{}_{l}^{m}(\kappa_n,\mathbf{r}) + b_{inc}^{(n)}(l,m)\mathbf{N}^{(r)}{}_{l}^{m}(\kappa_n,\mathbf{r}) \right], \tag{3}$$

$$\mathbf{E}_{int}^{(n)}(\mathbf{r}) = \sum_{\alpha} \left[a_{int}^{(n)}(l,m)\mathbf{M}^{(r)}{}_{l}^{m}(\kappa_n,\mathbf{r}) + b_{int}^{(n)}(l,m)\mathbf{N}^{(r)}{}_{l}^{m}(\kappa_n,\mathbf{r}) \right]. \tag{4}$$

On the other hand, the scattered field coefficients outside the particle can be expressed using singular multipoles:

$$\mathbf{E}_{sca}^{(n)}(\mathbf{r}) = \sum_{\alpha} \left[a_{sca}^{(n)}(l,m)\mathbf{M}^{(s)}{}_{l}^{m}(\kappa_n,\mathbf{r}) + b_{sca}^{(n)}(l,m)\mathbf{N}^{(s)}{}_{l}^{m}(\kappa_n,\mathbf{r}) \right], \tag{5}$$

in which $a_{\iota}^{(n)}(l,m)$ and $b_{\iota}^{(n)}(l,m)$ are the magnetic and electric multipole moments, respectively, measured in V/m. The regular (singular) magnetic and electric multipoles are defined as $\mathbf{M}^{(r)}{}_{l}^{m}(\kappa_n,\mathbf{r})$ $\left(\mathbf{M}^{(s)}{}_{l}^{m}(\kappa_n,\mathbf{r})\right)$ and $\mathbf{N}^{(r)}{}_{l}^{m}(\kappa_n,\mathbf{r})$ $\left(\mathbf{N}^{(s)}{}_{l}^{m}(\kappa_n,\mathbf{r})\right)$, respectively, where $\kappa_n = \dfrac{\omega - n\Omega}{c}$ is the wavenumber associated with each harmonic in free space. The degree and order of the multipoles are l and m, respectively, belonging to the integer sets $[1,+\infty]$ and $[-l,+l]$.

Inside the particle, the electromagnetic fields must satisfy the time-varying vector wave equation as shown below:

$$\nabla\times\nabla\times\left(\sum_{n=-\infty}^{+\infty}\mathbf{E}_{int}^{(n)}(\mathbf{r})e^{-i(\omega-n\Omega)t}\right) = \frac{\partial^2}{\partial t^2}\left(\epsilon_0\epsilon_r(t)\sum_{n=-\infty}^{+\infty}\mathbf{E}_{int}^{(n)}(\mathbf{r})e^{-i(\omega-n\Omega)t}\right), \tag{6}$$

which by expanding $\epsilon_r(t)$ in terms of a Fourier series with coefficients $c_n = c_{-n}^{*}$, and utilizing the inherent properties of the multipoles, results in the following eigenvalue equation to describe the dynamic medium:

$$\frac{(\omega-n\Omega)^2}{c^2}\sum_{n'=-\infty}^{+\infty} c_{n-n'}a_{int}^{(n')}(l,m) = \kappa^2 a_{int}^{(n)}(l,m), \qquad n=-\infty,\ldots,+\infty \tag{7}$$

where $c = \dfrac{1}{\sqrt{\mu_0\epsilon_0}}$ is the speed of light in a vacuum. By truncating the summation over n to N_n, an eigenvalue problem in the form $\mathbf{Au} = \kappa^2\mathbf{u}$ emerges. Here, $\mathbf{A}$ is a square matrix of rank $2N_n+1$ with its elements given in Eq. (7), κ is the eigenvalue or the wavenumber in the dynamic medium, and $\mathbf{u}$ is its associated eigenvector. For conceptual clarity, this study considers a dispersionless material to focus on the fundamental mechanisms of quantum emitter control. However, in practical scenarios, material dispersion and loss play a crucial role and can be incorporated by modifying the eigenvalue equation governing the medium. Dispersion in time-modulated meta-atoms can give rise to unique light-matter interaction phenomena, which is fully addressed in [90].

Having obtained the eigenvalues and eigenvectors of the time-varying material, the internal field coefficients can be spectrally expanded as follows:

$$\mathbf{E}_{int}^{(n)}(\mathbf{r}) = \sum_{u=-N_n}^{+N_n}\sum_{\alpha}\left[a_{int}^{(u)}(l,m)u_u^{(n)}\mathbf{M}^{(r)}{}_{l}^{m}(\kappa_u,\mathbf{r}) + b_{int}^{(u)}(l,m)u_u^{(n)}\mathbf{N}^{(r)}{}_{l}^{m}(\kappa_u,\mathbf{r})\right], \tag{8}$$

which implies that the electric field inside the particle is a superposition of waves with all possible wavenumbers. By truncating the summation over l to N_l and imposing the boundary conditions on the particle's surface, the multipole moments of the internal and scattered fields can be related to those of the incident field as follows:

$$TE:\begin{cases} a_{int}^{(u)}(l,m) = I_{TE}^{(p,u)}(l,m)a_{inc}^{(p)}(l,m), \\ a_{sca}^{(n)}(l,m) = T_{TE}^{(p,n)}(l,m)a_{inc}^{(p)}(l,m), \end{cases} \qquad TM:\begin{cases} b_{int}^{(u)}(l,m) = I_{TM}^{(p,u)}(l,m)b_{inc}^{(p)}(l,m), \\ b_{sca}^{(n)}(l,m) = T_{TM}^{(p,n)}(l,m)b_{inc}^{(p)}(l,m), \end{cases} \tag{9}$$

in which $I_{TE}^{(p,u)}(l,m)$ $\left(T_{TE}^{(p,n)}(l,m)\right)$ and $I_{TM}^{(p,u)}(l,m)$ $\left(T_{TM}^{(p,n)}(l,m)\right)$ are the dynamic Mie coefficients that connect the magnetic and electric multipole moments of the p-th harmonic of the incident field to those of the internal (external) field with the harmonic order u (n), respectively. These coefficients can be used to model the interaction of a cluster of time-modulated particles with light (refer to the Supplemental Material for a detailed derivation of the dynamic Mie coefficients [88]).

2.2 Mutual interactions of the particles in a cluster

Considering a random distribution of M time-varying resonators with their centers at $\mathbf{r}_c = [\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_M]$, the total incident field moments in the local coordinates of the j-th particle can be calculated by accounting for both the external excitation and scattered fields from the other resonators. This objective can be achieved for the magnetic and electric moments by changing the basis as follows [91, 92]:

$$ {}^{j}a_{inc,t}^{(p)}(l,m) = {}^{j}a_{inc}^{(p)}(l,m) + \sum_{k=1}^{M(k\neq j)} \sum_{\beta\in\{\nu,\mu\}} \left[{}^{p}A_{l\nu}^{m\mu}(\mathbf{r}_{jk})^{k}a_{sca}^{(p)}(\nu,\mu) + {}^{n}B_{l\nu}^{m\mu}(\mathbf{r}_{jk})^{k}b_{sca}^{(p)}(\nu,\mu) \right], \tag{10a} $$

$$ {}^{j}b_{inc,t}^{(p)}(l,m) = {}^{j}b_{inc}^{(p)}(l,m) + \sum_{k=1}^{M} \sum_{\beta} \left[{}^{p}B_{l\nu}^{m\mu}(\mathbf{r}_{jk})^{k}a_{sca}^{(p)}(l,m) + {}^{p}B_{l\nu}^{m\mu}(\mathbf{r}_{jk})^{k}b_{sca}^{(p)}(l,m) \right], \tag{10b} $$

in which ν and μ are the translation indices varying within the integer sets $[1, N_l]$ and $[-\nu, +\nu]$, respectively, and $\mathbf{r}_{jk} = \mathbf{r}_j - \mathbf{r}_k$ is the separation vector between the particles. The translation coefficients are defined as:

$$ {}^{n}A_{l\nu}^{m\mu}(\mathbf{r}_{jk}) = \langle \mathbf{M}^{(s)}{}_{l}^{m}(\kappa_n, {}^{k}\mathbf{r}) \big| \mathbf{M}^{(r)}{}_{\nu}^{\mu}(\kappa_n, {}^{j}\mathbf{r}) \rangle = \langle \mathbf{N}^{(s)}{}_{l}^{m}(\kappa_n, {}^{k}\mathbf{r}) \big| \mathbf{N}^{(r)}{}_{\nu}^{\mu}(\kappa_n, {}^{j}\mathbf{r}) \rangle , \tag{11a} $$

$$ {}^{n}B_{l\nu}^{m\mu}(\mathbf{r}_{jk}) = \langle \mathbf{M}^{(s)}{}_{l}^{m}(\kappa_n, {}^{k}\mathbf{r}) \big| \mathbf{N}^{(r)}{}_{\nu}^{\mu}(\kappa_n, {}^{j}\mathbf{r}) \rangle = \langle \mathbf{N}^{(s)}{}_{l}^{m}(\kappa_n, {}^{k}\mathbf{r}) \big| \mathbf{M}^{(r)}{}_{\nu}^{\mu}(\kappa_n, {}^{j}\mathbf{r}) \rangle , \tag{11b} $$

where $\langle * | * \rangle$ is an inner product defined on the surface of a sphere of radius $|\mathbf{r}_{jk}|$ centered at $\mathbf{r}_k$, and ${}^{j}\mathbf{r}$ and ${}^{k}\mathbf{r}$ are measured from the centers of the j-th and k-th particles, respectively.
By substituting the total incident field of each resonator into Eq. (7), a coupled system of linearly independent equations in the form $\mathbf{T}\mathbf{x} = \mathbf{y}$ can be obtained, where $\mathbf{T}$ is the interaction matrix given below:

$$ \mathbf{T} = \begin{bmatrix} (\mathbf{T}_1)^{-1} & \mathbf{T}_{12} & \ldots & \mathbf{T}_{1M} \\ \mathbf{T}_{21} & (\mathbf{T}_2)^{-1} & \ldots & \mathbf{T}_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{T}_{M1} & \mathbf{T}_{M2} & \ldots & (\mathbf{T}_M)^{-1} \end{bmatrix}, \tag{12} $$

in which the diagonal blocks should be obtained using Eq. (9), while Eq. (10) should be used to calculate the off-diagonal elements. The unknown vector is defined as $\mathbf{x} = [\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \ldots, \tilde{\mathbf{x}}_M]^{\dagger}$, with:

$$ \tilde{\mathbf{x}}_j = \left[{}^{j}\mathbf{a}_{sca}, {}^{j}\mathbf{b}_{sca} \right]^{\dagger}, \tag{13} $$

and the excitation vector is described as $\mathbf{y} = [\tilde{\mathbf{y}}_1, \tilde{\mathbf{y}}_2, \ldots, \tilde{\mathbf{y}}_M]^{\dagger}$, with:

$$ \tilde{\mathbf{y}}_j = \left[{}^{j}\mathbf{a}_{inc}, {}^{j}\mathbf{b}_{inc} \right]^{\dagger}. \tag{14} $$

The obtained multipole moments can, therefore, be used to calculate various observable scattering characteristics (for a detailed description of the multipole translation coefficients and the elements of the interaction matrix, see the Supplemental Material [88]).

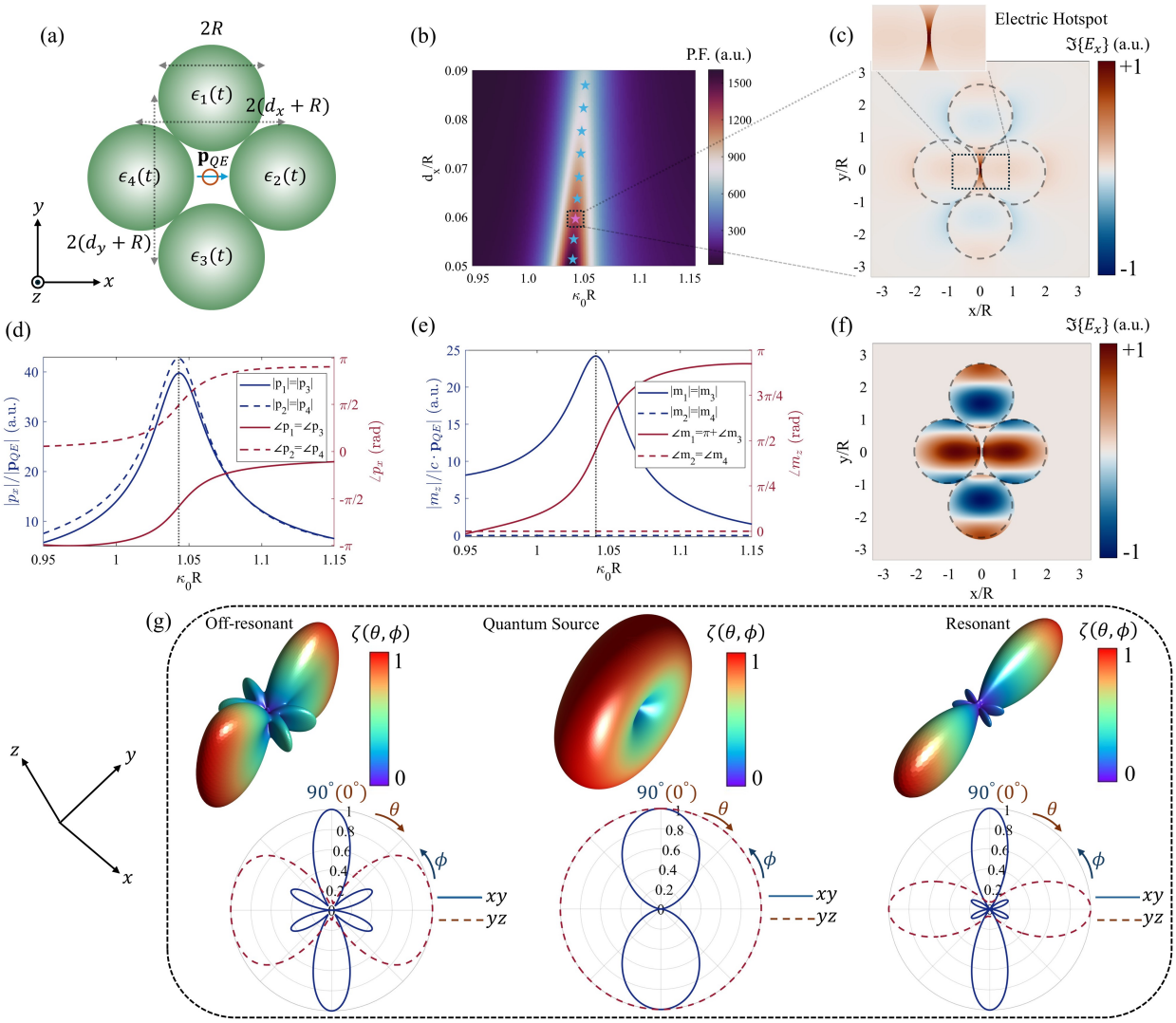


Figure 2: (a) Cross-section of the quantum emitter embedded in the ensemble of time-modulated Mie resonators and the associated design parameters. (b) Purcell factor as a function of frequency and distance on the x-axis. (c) Electric field along the quantum source's dipole moment at the resonant frequency and the corresponding electric hotspot formed at the center. (d) Electric and (e) magnetic dipole moments induced within the particles. (f) Electric field of the excited multipolar modes inside the resonators along the polarization direction of the quantum emitter. (g) Far-field radiation patterns of the optical system in the resonant and off-resonant conditions, and for the isolated quantum source. This figure shines a light on the design process of the static optical system.

2.3 Observable scattering characteristics of the ensemble of resonators

The quantum emitter in this paper is modeled as a two-level system, positioned at $\mathbf{r}_{\mathrm{QE}}$, with ground and excited states, denoted as $|g\rangle$ and $|e\rangle$, respectively. These states are associated with the energy levels of E_g and E_e, and the transition energy of the two-level system is given by $\hbar\omega_0 = E_e - E_g$, where ω_0 represents the transition angular frequency. Under the dipolar approximation, which neglects the contributions of higher-order multipoles, the interaction Hamiltonian for the hybrid field-atom system is expressed as [93, 94]:

$$\widehat{H}_I = -\widehat{\mathbf{p}} \cdot \widehat{\mathbf{E}}, \tag{15}$$

where $\widehat{\mathbf{E}}$ represents the vacuum electric field operator, and $\widehat{\mathbf{p}} = \mathbf{p}_{\mathrm{QE}} \left[|e\rangle \langle g| + |g\rangle \langle e| \right]$ denotes the dipole moment operator. The transition dipole moment, without loss of generality, is defined as a real quantity $\mathbf{p}_{\mathrm{QE}} = \langle g|\widehat{\mathbf{p}}|e\rangle = \langle e|\widehat{\mathbf{p}}|g\rangle$.
Utilizing Fermi's golden rule and after appropriate mathematical manipulations, the spontaneous emission decay rate of the quantum emitter is given by [95, 96]:

$$\gamma = \frac{\pi\omega_0}{\hbar\epsilon_0} |\mathbf{p}_{\mathrm{QE}}|^2 \sum_{\mathbf{k}} \left[\frac{\mathbf{p}_{\mathrm{QE}}}{|\mathbf{p}_{\mathrm{QE}}|} \cdot (\mathbf{u}_{\mathbf{k}} \mathbf{u}_{\mathbf{k}}^*) \cdot \frac{\mathbf{p}_{\mathrm{QE}}}{|\mathbf{p}_{\mathrm{QE}}|} \right] \delta(\omega_{\mathbf{k}} - \omega_0), \tag{16}$$

where $\mathbf{u}_{\mathbf{k}}$ are the normalized eigenmodes of photons that satisfy the vector wave equation:

$$\nabla \times \nabla \times \mathbf{u}_{\mathbf{k}}(\mathbf{r}, \omega_{\mathbf{k}}) - \frac{\omega_{\mathbf{k}}^2}{c^2} \mathbf{u}_{\mathbf{k}}(\mathbf{r}, \omega_{\mathbf{k}}) = 0. \tag{17}$$

Since the dyadic Green's function satisfies the vector wave equation with a Dirac delta function excitation, it can be expanded in terms of the normal eigenmodes as:

$$\Im \left\{ \overline{\overline{\mathbf{G}}}(\mathbf{r}_{\mathrm{QE}}, \mathbf{r}_{\mathrm{QE}}; \omega) \right\} = \frac{\pi c^2}{2\omega} \sum_{\mathbf{k}} \mathbf{u}_{\mathbf{k}}^*(\mathbf{r}_{\mathrm{QE}}, \omega_{\mathbf{k}}) \mathbf{u}_{\mathbf{k}}(\mathbf{r}_{\mathrm{QE}}, \omega_{\mathbf{k}}) \delta(\omega_{\mathbf{k}} - \omega_0). \tag{18}$$

Thus, the Purcell factor, which characterizes the modified spontaneous emission rate due to the electromagnetic environment, is given by [97]:

$$PF = \frac{\gamma}{\gamma_0} = 1 + \frac{6\pi}{\kappa} \left[\frac{\mathbf{p}_{\mathrm{QE}}}{|\mathbf{p}_{\mathrm{QE}}|} \cdot \Im \left\{ \overline{\overline{\mathbf{G}}}(\mathbf{r}_{\mathrm{QE}}, \mathbf{r}_{\mathrm{QE}}; \omega_0) \right\} \cdot \frac{\mathbf{p}_{\mathrm{QE}}}{|\mathbf{p}_{\mathrm{QE}}|} \right] = 1 + \frac{6\pi\epsilon_0}{|\mathbf{p}_{\mathrm{QE}}|^2} \frac{1}{\kappa^3} \Im \left\{ \mathbf{p}_{\mathrm{QE}} \cdot \mathbf{E}_s(\mathbf{r}_{\mathrm{QE}}) \right\}, \tag{19}$$

where $\overline{\overline{\mathbf{G}}}_s$ and $\mathbf{E}_s$ represent the secondary parts of the dyadic Green's function and the electric field, respectively, arising from the inhomogeneous background.
Hence, within the dipolar approximation and weak coupling regime, the quantum emitter's response is fully determined by the Green's function of the inhomogeneous medium, computed here using a multipole expansion. Herein, observables are calculated using a classical dipole moment and normalized to its free-space response to establish an equivalence between the quantum-mechanical picture of spontaneous emission and classical dipole radiation by resorting to the dyadic Green's function of the inhomogeneous environment [93, 94] (a detailed derivation of the transition from quantum formalism to classical electrodynamics can be found in the Supplemental Material [88]).
The scattering cross section measured at each harmonic frequency can be obtained by integrating the time-averaged Poynting vector, which can be expanded in terms of the multipole moments as given below:

$$\sigma_{sca}^{(n)} = \frac{1}{P_0} \frac{2\pi}{\kappa_n(\omega - n\Omega)\,\mu_0} \sum_{l=1}^{N_l} \left[\left| a_{sca}^{(n)}(l,m) + a_{inc}^{(n)}(l,m) \right|^2 + \left| b_{sca}^{(n)}(l,m) + b_{inc}^{(n)}(l,m) \right|^2 \right], \tag{20}$$

in which all the multipole moments are translated to the corresponding coordinates of the quantum emitter. It should be noted that $P_0 = \frac{c\kappa_0^4}{12\pi\epsilon_0} \left|\mathbf{p}_{\mathrm{QE}}\right|^2$ is the radiated power of the equivalent dipole in free space [98, 99]. Finally, the radiation pattern in the far-field region for each harmonic is calculated as:

$$\zeta^{(n)}(\theta, \phi) = \frac{1}{P_0} \frac{2\pi}{\eta_0} r^2 \left| \mathbf{E}_{sca}^{(n)} + \mathbf{E}_{inc}^{(n)} \right|^2, \tag{21}$$

where $\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$ is the wave impedance of free space. It is worth noting that the electromagnetic fields of the quantum emitter are expressed in terms of singular multipoles in its local coordinates (see the Supplemental Material for the derivation of the multipole moments for the quantum emitter's radiation [88]). Finally, although the quantum emitter is modeled as a point source, if its dimensions are small enough for the interaction Hamiltonian to be well approximated by the dipolar term, it effectively behaves as a finite-sized emitter whose spectral and spatial attributes can be controlled using time-varying Mie resonators.

3 Quantum emitter interaction with static resonators

Before embarking on the dynamic control of the quantum emitter, this section delves into the design procedure of the structure and the physics underlying its performance. As depicted in Figure 2a, the proposed structure comprises four identical spherical particles with radius R, symmetrically distributed in the xy-plane. The separation on the $x-$axis is denoted as d_x, and its counterpart on the y-axis is chosen to be $d_y = 2\sqrt{4R^2 - (\frac{d_x}{2} + R)^2}$, keeping the resonators in close proximity to each other and ensures the maximum coupling. The quantum emitter is modeled as a light-emitting source with an angular frequency of ω and an electric dipole moment of $\mathbf{p}_{\mathrm{QE}} = p_0 \hat{a}_x$, positioned at the origin of the coordinates. For the rest of this section, the particles are considered to be silicon with a static relative permittivity of $\epsilon_r = 11.56$, and the emphasis is placed on optimizing the geometrical configuration of the system. Besides, all the results are presented with respect to the dimensionless size parameter $\kappa_0 R = \frac{\omega}{c} R$ for the sake of scalability. It is also worth noting that the results in this manuscript are obtained with at least $N_l = 12$ to ensure convergence (refer to the Supplemental Material for the convergence test and numerical validation of the results [88]).
Due to the particles' considerably high permittivity, they can support multiple magnetic and electric resonant modes. Consequently, the interplay between the excited multipole modes within the resonators can be manipulated to enhance the spontaneous emission rate of the quantum emitter, which can be quantitatively captured by resorting to the Purcell factor. Figure 2b illustrates the Purcell factor as a function of the distance on the x-axis (d_x) for various size parameters. As it can be clearly seen, the distance can be adjusted to achieve a sharp and robust electric field resonance, where the Purcell factor reaches its maximum. It should be noted that there is a trade-off between the resonance strength and its sensitivity to the separation. Smaller values of d_x result in larger Purcell factors but also lead to significantly higher sensitivity to spatial changes. In this context, the distance on the x-axis is set to $\frac{d_x}{R} = \frac{1}{15}$, which results in a resilient resonance at $\kappa_r R = 1.046$ with a remarkably high Purcell factor ($PF \approx 990$). To further elaborate on the system's performance under resonant conditions, Figure 2c shows the imaginary part of the electric field in the direction of the quantum emitter. As suggested by Eq. (15), the Purcell factor is directly connected to the electric field concentration at the quantum emitter's location, which is also proportional to the local density of electromagnetic states in the classical picture. Therefore, the electric hotspot formed at the origin, as illustrated in Figure 2c, is in charge of the intensified Purcell factor, implying the enhanced emission rate of the quantum emitter.
In order to shine a light on the origin of the excited modes inside the resonators, the electric and magnetic dipole moments (magnitude and phase are shown in blue and red, respectively) induced within the particles are depicted in Figures 2e and 2f, respectively. It should be noted that these are the only predominant excited multipoles in the system, and other higher-order resonant modes in the local coordinates of the particles can be neglected without diminishing the accuracy (a detailed analysis of higher-order modes is provided in the Supplemental Material [88]). Due to the inherent symmetry of the designed static system and the polarization of the quantum emitter, the electric dipole moments induced in the resonators on the x-axis (particles one and three) are identical. Similarly, the particles aligned on the y-axis (particles two and four) have the same generated electric dipole moments. As shown in Figure 2d, the amplitude of the electric dipole moment in all the resonators reaches its peak at resonance. Additionally, the electric dipole moments in particles one and three have a π phase difference compared to those in particles two and four. Besides, considering the problem's symmetry in the presence of the quantum emitter, only the particles on the y-axis (particles one and three) experience an out-of-plane (z-directed) magnetic dipole moment, while

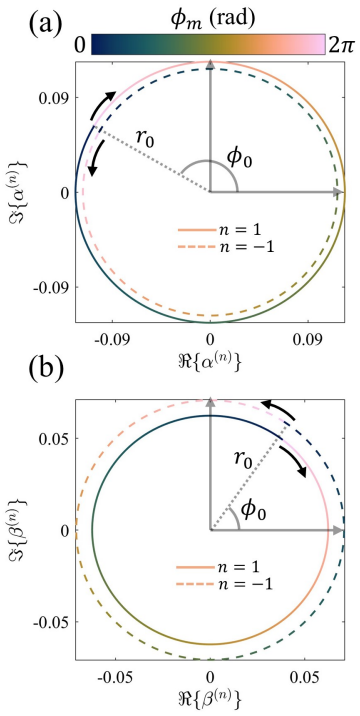


Figure 3: Dynamics of the dipolar (a) magnetic and (b) electric Mie coefficients of an isolated resonator associated with the harmonic $n = \pm 1$ with respect to the modulation phase, while keeping the modulation strength constant. Variations in color indicate changes in the modulation phase, and each point on the circles corresponds to the relative value of the dynamic magnetic or electric Mie coefficients compared to their static counterparts. These trajectories illustrate the tuning mechanism for the dynamic control of the quantum emitter.

the excited modes within the particles on the x-axis are solely of an electric nature. As illustrated in Figure 2e, despite the out-of-phase behavior of the resonances caused by the quantum emitter's imposed magnetic moment, the induced magnetic dipole moments inside particles one and three have the same magnitude. Therefore, the resonance in the proposed structure is of hybrid origin, including both electric and magnetic dipole modes. The electric field inside the resonators, shown in Figure 2f, clearly illustrates the structure of the resonant modes within the particles and the system's mirror symmetry, which results in unique radiation characteristics.

Figure 2g demonstrates the radiation pattern of the system in three scenarios. In the absence of the resonators, the quantum emitter behaves like an incoherent light source (a spatially incoherent source of light that contains a complex profile in the wavevector domain and cannot be described using a single wavevector) with a torus-shaped radiation pattern. On the contrary, the light-matter interaction with the particles results in considerably directive radiation in both the xy- and yz- planes. Figure 2g also shows that the transition from a resonant to an off-resonant state dissipates the ensemble's radiation directivity. The mirror symmetry in the near-field region translates to symmetric radiation in the far-field, which can be skillfully manipulated through time modulation to achieve the desired scattering characteristics. The structure discussed in this section will serve as the baseline for the dynamic control of the quantum emitter.

4 Dynamic control of the quantum emitter

In this section, time modulation, introduced as the periodic variation of the particles' relative permittivities in time, is used to dynamically manipulate the system's response to the quantum emitter. The dynamics

of the Mie coefficients associated with the dipolar magnetic and electric modes are primarily investigated for the prominent harmonics in the case of a sinusoidal modulation function. It is then demonstrated how the interaction of the resonators can be controlled, using time modulation as a closed feedback system, to compensate for frequency detuning of the quantum source. Additionally, the far-field pattern of the structure is directed to various angles in the xy-plane, and it is shown that the transverse Kerker conditions can be satisfied by adequately altering the amplitude and phase of the modulation for each particle. Finally, time modulation is examined as a switch that can transition the structure from a non-radiating anapole mode to a resonant radiating state.

4.1 Dynamics of the Mie coefficients in a time-varying resonator

After designing the static structure, the relative permittivity of the particles can be controlled over time to achieve the desired scattering characteristics. To elaborate on how time modulation can be exploited to manipulate the interaction of the resonators and achieve the intended objectives, the dynamics of the Mie coefficients related to the magnetic and electric dipolar modes in a single particle are explored for the predominant harmonics ($n = \pm 1$). The resonator is composed of a periodic time-varying permittivity defined as follows:

$$\epsilon_r(t) = \epsilon_{dc}\left(1 + \delta\cos\left(\Omega t + \phi_m\right)\right), \tag{22}$$

in which $\epsilon_{dc} = 11.56$ denotes the static part of the relative permittivity, $\delta = 0.1$ stands for the modulation strength, $\Omega = 0.01\omega$ shows the modulation angular frequency, and ϕ_m is the modulation phase. Even though the permittivity of each particle contains two degrees of freedom (modulation strength and phase), the emphasis here is solely placed on the modulation phase, as it can be utilized to manipulate the relative phase between the Mie coefficients in the time-varying and static scenarios. Therefore, the dimensionless variables $\alpha^{(n)} = \dfrac{T_{TE}^{(p,n)}(l,m)}{T_{TE}(l,m)}$ and $\beta^{(n)} = \dfrac{T_{TM}^{(p,n)}(l,m)}{T_{TM}(l,m)}$ ($p = 0$, $n = \pm 1$, and $l = 1$) are introduced to quantify the time modulation-induced phase on the dipolar magnetic and electric Mie coefficients, respectively. It is also worth noting that $T_{TE}(l,m)$ and $T_{TM}(l,m)$ are the magnetic and electric Mie coefficients of the static resonator.

The trajectories for $\alpha^{(n)}$ and $\beta^{(n)}$ under a varying modulation phase ϕ_m are illustrated in Figures 3a and 3b, respectively. As can be seen, for a constant modulation strength, the amplitudes of the magnetic and electric Mie coefficients remain unchanged, translating into circles with a radius of r_0 in the complex plane. Moreover, as also suggested by Eq. (7), the relative phase of both the magnetic and electric Mie coefficients follows the modulation phase with an offset of ϕ_0, rendering the static phase imposed by the resonator's inherent physical properties. It should be noted that the coefficients associated with the harmonic number $n = +1$ undergo counterclockwise rotational motion in the complex plane, while those of the harmonic with $n = -1$ tend to experience clockwise rotation, resembling the nature of the time-Floquet modes with opposite numbers. Additionally, as demonstrated in Figures 3a and 3b, due to the nonadiabatic nature of the variations in time, the Mie coefficients related to the harmonics with $n = +1$ and $n = -1$ no longer maintain the same amplitude and phase, suggesting possible energy exchange within the system [89, 90]. These principles can be applied to manipulate the radiation of the quantum emitter using time-varying Mie resonators, as will be discussed further in this section.

4.2 Compensation for frequency detuning in the quantum emitter's radiation

In order to ensure the optimal performance, not only the static cavity needs to be designed to perform in the resonant condition with $\kappa_r R = \dfrac{\omega_r}{c} R = 1.046$ but also the quantum source is expected to radiate photons at the resonant frequency. Nevertheless, because of the stochastic nature of the quantum source in terms of its size, shape, material properties, and host medium, it might radiate at a frequency slightly different from the intended resonant frequency of the cavity [100, 101], which is also referred to as frequency detuning and defined as $\Delta\kappa_0 R = \dfrac{\omega - \omega_r}{\omega_r}$ in this manuscript. As previously explained, time modulation can be introduced to dielectric resonators in order to dynamically manipulate the amplitude and phase of the Mie coefficients, thereby opening up the possibility of controlling the scattering characteristics of the quantum emitter. Additionally, as suggested by Eq. (2), the wave scattered from a time-modulated optical system is

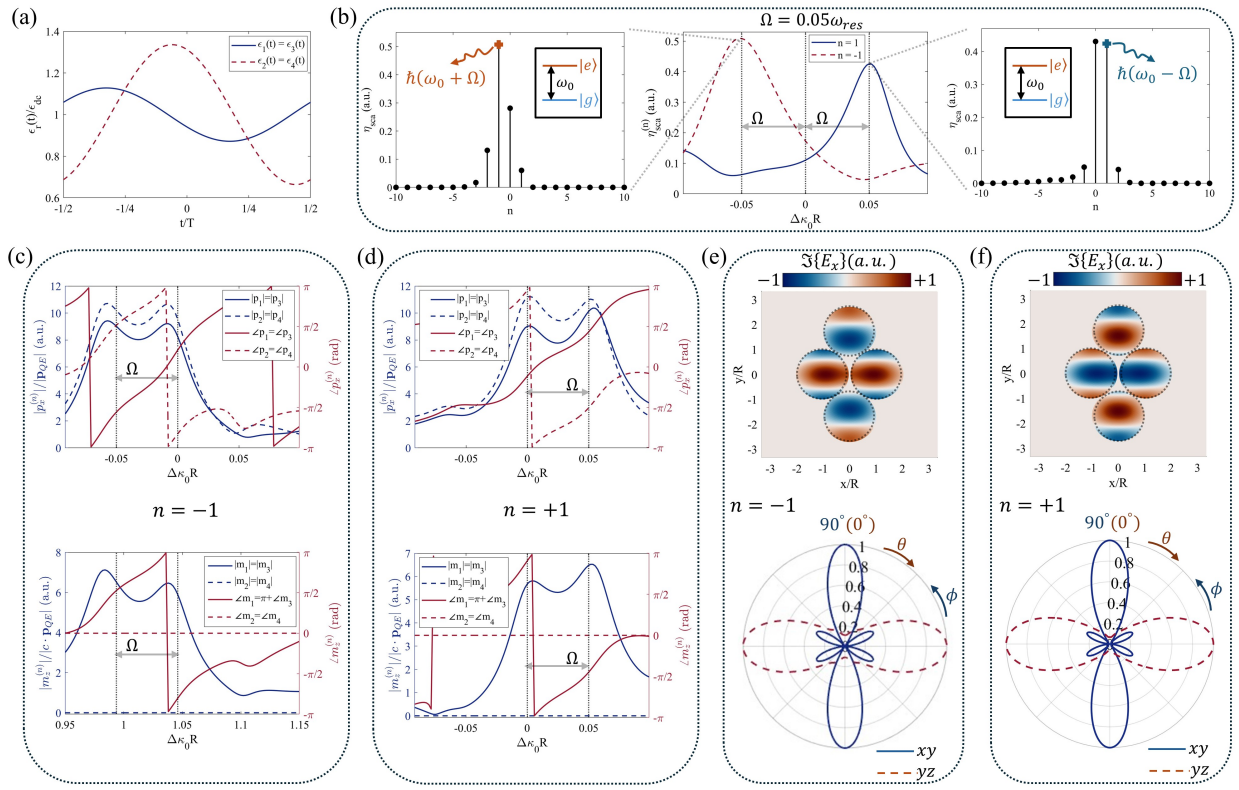


Figure 4: (a) Modulation waveforms of the resonators optimized to compensate for frequency detuning of the quantum source. (b) Scattering efficiency curves for the harmonics with $n = \pm 1$ as a function of detuning frequency, and the Fourier coefficients of the scattering efficiency calculated at $\Delta\kappa_0 R = \pm 0.05$. Electric and magnetic dipole moments excited inside the particles associated with the harmonics with (c) $n = -1$ and (d) $n = +1$. Electric fields along the polarization direction of the quantum emitter and polar far-field patterns at $\Delta\kappa_0 R = -0.05$ and $\Delta\kappa_0 R = +0.05$ for the harmonics with (e) $n = -1$ and (f) $n = +1$, respectively. This figure demonstrates the feedback strategy based on time modulation to compensate for frequency detuning in the quantum emitter's radiation.

no longer monochromatic and contains various harmonics of the modulation frequency. This phenomenon is also known as the frequency-mixing property of time modulation [64, 65]. Therefore, subject to the adequate choice of both the modulation frequency and its waveform, time modulation can be exploited to compensate for the frequency detuning of the quantum emitter by translating its contents to the desired harmonic, which coincides with the cavity's resonant frequency.

The approach implemented in this study to compensate for the frequency detuning of the quantum emitter is based on an optimal feedback strategy. This strategy measures the frequency detuning, sets it equal to the modulation frequency, and optimizes the modulation functions of the particles to achieve the highest scattering efficiency at harmonics with $n = +1$ or $n = -1$, depending on the sign of frequency detuning. It is also worth noting that the scattering efficiency, $\eta_{sca}^{(n)} = \dfrac{\sigma_{sca}^{(n)}}{\sum_{n'} \sigma_{sca}^{(n')}}$, is a dimensionless parameter defined to quantify the power transferred to the n-th harmonic. In order to shine a light on the aforementioned approach, the feedback strategy is harnessed to atone for frequency detuning in a quantum emitter which radiates at $\omega_0 = \omega_r\,(1 \pm 0.05)$. The angular modulation frequency is chosen to be $\Omega = 0.05\omega_r$, and the modulation functions for the particles along the same axis are considered identical to preserve the system's symmetry. The presence of symmetry ensures that the excited multipoles are of the same parity as in the

static case, resulting in the desired far-field radiation patterns [102, 103]. It should also be noted that all optimizations are performed using a hybrid platform based on a genetic algorithm and interior-point method (further details can be found in the Supplemental Material [88]).
The optimal modulation waveforms to compensate for frequency detuning and the corresponding scattering efficiencies are illustrated in Figures 4a and 4b, respectively. As can be seen, the scattering efficiency for the harmonic with $n = +1$ ($n = -1$) reaches its maximum at $\Delta\kappa_0 R = +0.05$ ($\Delta\kappa_0 R = -0.05$), indicating that the optical system radiates photons with energy $\hbar\omega_r$ at nearly 50% efficiency. It should be noted that the sinusoidal modulation functions can be replaced with periodic signals comprising more Fourier coefficients. These Fourier coefficients can subsequently be optimized to acquire near-unity scattering efficiency, which its detail is omitted in the main manuscript for the sake of brevity (refer to the Supplemental Material to see the examples [88]). In order to shed light on the performance of the resonators, the electric and magnetic dipole moments induced within the particles at the harmonics with $n = -1$ and $n = +1$ are depicted in Figures 4c and 4d, respectively. As evident from Figure 4c, following the symmetry of the ensemble, the excited electric dipole moments within the particles along the same axis are identical and oriented in the x-direction, parallel to the polarization direction of the quantum source. However, as anticipated, only the particles on the y-axis experience a magnetic dipole, while the excited modes within the other particles are purely electric. Furthermore, besides the resonance at zero detuning, which corresponds to the static resonant frequency of the cavity, both the electric and magnetic dipole moments exhibit an additional resonance near the detuning frequency. This additional resonance is responsible for the strong energy transfer to the desired harmonic, thereby compensating for the detuning of the quantum emitter. On the other hand, as shown in Figure 4d, for the harmonic with $n = +1$, the electric and magnetic dipole moments exhibit a similar behavior to those with $n = -1$, except that the additional resonance is located on the positive side, corresponding to a positive detuning, to atone for the frequency detuning in the quantum emitter's radiation.
Finally, the electric fields inside the particles and their polar radiation patterns are given in Figures 4e and 4f for the harmonics with $n = -1$ and $n = +1$, respectively. As can be seen, the excited modes within the particles in both scenarios exhibit a similar electric field pattern to the static case, indicating a resonant radiating state of the resonators. Moreover, the electric fields for both harmonics preserve the mirror symmetry in the xy-plane, which is crucial for achieving the desired far-field radiation pattern. However, there is a π phase shift between the electric fields for $n = -1$ and $n = +1$, reflecting the inherent difference in the time-Floquet modes with opposite signs. Furthermore, considering the symmetry of the resonators, the far-field radiation patterns closely resemble the static case. This demonstrates that the proposed feedback strategy based on time modulation can not only shift the resonance to the desired frequency but also recover its radiation characteristics, which is crucial in numerous on-chip applications. However, it should be noted that this method compensates for frequency detuning by dynamically tuning the resonator's spectral response, but it does not alter the quantum state or coherence properties of the emitter.

4.3 Directional control of the quantum emitter and transverse Kerker effect

In addition to compensating for frequency detuning, controlling the directional radiation of the quantum emitter is crucial for various applications in advanced integrated quantum physics. As mentioned earlier, in the absence of the cavity, the quantum emitter behaves as an incoherent light source with a torus-shaped radiation pattern. Integrating the quantum emitter with the resonators can significantly enhance directivity, resulting in a far-field radiation pattern with two main lobes in the xy- and yz-planes. Nonetheless, the existing symmetry of the problem constrains the excited multipole moments within the particles, thereby limiting the achievable radiation pattern in the static configuration. By adding time as a new variable to the design paradigm, the symmetry can be broken to induce the desired multipoles in the particles. Besides, the mutual interactions between the resonators can be deliberately engineered through the careful selection of modulation functions. In this context, assuming the quantum source radiates at $\omega_0 = \omega_r$, time modulation is exploited to dynamically steer the radiation direction of photons with energy $\hbar(\omega_0 - \Omega)$, associated with the harmonic number $n = +1$, to the desired angles in the xy-plane. It should be noted that the angular modulation frequency is considered to be $\Omega = 0.01\omega_r$, and the mutual interactions between the particles are skillfully manipulated to attain maximum directivity toward the desired angles.
Herein, the modulation waveforms within the resonators are optimized to steer one of the main lobes of the radiation pattern, related to the harmonic with $n = +1$, to the intended angles in the xy-plane (ϕ) while

suppressing the other lobe. The radiation patterns for various steering angles in the xy-plane, along with their associated modulation parameters for different particles, are depicted in Figures 5a and 5b, respectively. It is also worth noting that only the parameters for the angles in the first quadrant are required to span the entire xy-plane, since the other directions can be readily obtained by leveraging the symmetry of the geometrical configuration of the resonators. To further elucidate the system's performance, the modulation functions, scattering efficiencies for different harmonics, and the radiation pattern of the photons for $\phi = 90^\circ$ and $n = +1$ are shown in Figure 5c. As can be seen, the symmetry is broken on the y-axis, directing the photons toward $\phi = 90^\circ$. Additionally, to better comprehend the working principles of the system, the electric and magnetic dipole moments induced in the particles for the harmonic with $n = +1$ are demonstrated in Figure 5d. As evident from the broken symmetry along the y-axis, the induced electric dipole moments in particles one and three are no longer identical, and the particles aligned along the x-axis also experience magnetic modes.

Figure 5e illustrates the forward scattering cross section and forward-to-backward ratio as a function of frequency and steering angle. The colormap corresponds to the forward scattering cross section, and the forward-to-backward ratio is depicted using white contours (the main and side radiation lobes are marked with stars). Additionally, a cross-sectional view of these values and the backward scattering cross section at the resonant frequency is included. As a direct consequence of the broken symmetry due to time modulation, as demonstrated in Figure 5e, in the vicinity of the resonant frequency of the cavity, both the forward scattering cross section and the forward-to-backward ratio peak at $\phi = 90^\circ$. This indicates that the radiation has one main lobe in the desired direction while the other lobe is successfully suppressed. It should be noted that at the resonant frequency, the other two peaks in the forward-to-backward ratio, symmetrically distributed around $\phi = 90^\circ$, correspond to the side lobes and can be overlooked. In fact, although the backward scattering cross section tends to zero, its counterpart in the forward direction is also negligible compared to the main radiation lobe. Furthermore, the electric field inside the particles, along with the polar radiation pattern of the system in the xy- and yz-planes for the harmonic with $n = +1$, are illustrated in Figure 5f. Clearly, both the excited modes within the particles and the associated radiation patterns adhere to the broken symmetry of the problem, resulting in a single radiation peak at the desired angle.

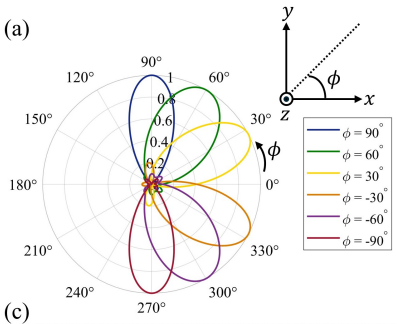
Besides steering the radiation toward the desired angles, the interaction between the excited multipolar modes within the resonators can be precisely engineered to satisfy transverse Kerker conditions for the quantum emitter [47]. In the absence of the resonators, the radiation of the quantum emitter is not only torus-shaped but also does not radiate any power in its polarization direction. By resorting to the time-modulated cavity, transverse Kerker conditions in the xy-plane can be fulfilled, rotating the main radiation lobes and aligning them with the polarization direction of the quantum emitter. The conditions defined in this work can be summarized as follows:

$$\zeta(\theta,\phi)\Big|_{\theta=0^\circ,180^\circ} \approx 0, \tag{23a}$$

$$\zeta(\theta,\phi)\Big|_{\theta=90^\circ}^{\phi=90^\circ,270^\circ} \approx 0. \tag{23b}$$

These conditions ensure that radiation is mostly focused along the polarization of the quantum source. Careful selection of the modulation waveforms in the particles can result in a configuration with broken symmetry, allowing for the excitation of the desired multipolar modes to satisfy the aforementioned conditions. Therefore, the optimal modulation functions to satisfy transverse Kerker conditions for photons with energy $\hbar(\omega_0 - \Omega)$ and the corresponding scattering efficiencies for different harmonics are depicted in Figure 6a. It should be noted that the system is excited at $\omega_0 = \omega_r$, and the modulation angular frequency is assumed to be $\Omega = 0.01\omega_r$ in this scenario. The electric dipole along the x-axis and magnetic dipole moments induced within the particles for the harmonic with $n = +1$ are illustrated in Figure 6b. As can be seen, the excited modes are no longer symmetric, giving rise to electric dipole moments that are perpendicularly aligned with the dipole moment of the quantum emitter (y-axis). In this context, the excited electric dipole moments in the y-direction and the electric field inside the particles are demonstrated in Figure 6c. Clearly, the electric dipole moments along the y-axis appear only in the particles aligned on the x-axis with equal amplitudes and a π phase shift. This results in the asymmetric electric field in the ensemble, breaking the mirror symmetry. Finally, the radiation pattern of the system is shown in Figure 6d. It can be observed that by satisfying transverse Kerker conditions, the radiation is directed along the polarization of the quantum emitter in both

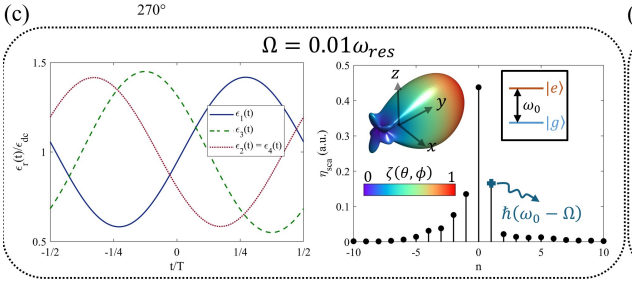
(a)



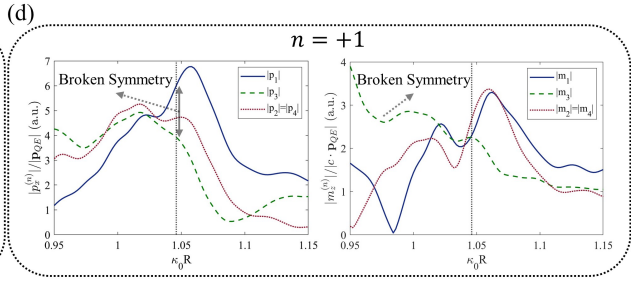
(b)

Target Angle	$\phi = 90^{\circ}$	$\phi = 60^{\circ}$	$\phi = 30^{\circ}$
$(\delta_1,\ \phi_{m1})$	$(0.42, 4.57)$	$(0.26, 2.41)$	$(0.16, 0.43)$
$(\delta_2,\ \phi_{m2})$	$(0.45, 0.79)$	$(0.47, 5.28)$	$(0.25, 4.10)$
$(\delta_3,\ \phi_{m3})$	$(0.42, 2.06)$	$(0.36, 6.10)$	$(0.21, 6.02)$
$(\delta_4,\ \phi_{m4})$	$(0.42, 2.06)$	$(0.41, 2.17)$	$(0.22, 1.38)$

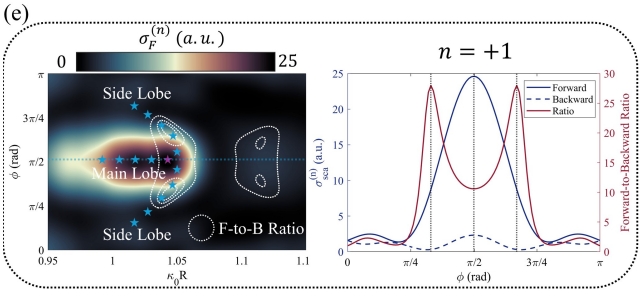
(c)



(d)



(e)



(f)

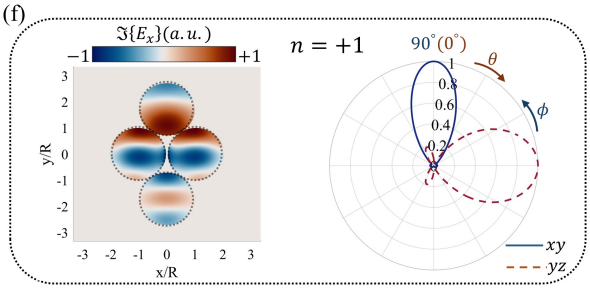


Figure 5: (a) Steering of the quantum source's radiation toward the desired angles in the xy-plane, and (b) the associated modulation parameters for the angles in the first quadrant. (c) Modulation functions of the particles optimized to align the beam with $\phi = 90°$, as well as the Fourier coefficients of the scattering efficiency and the radiation pattern of the harmonic with $n = +1$ at the resonant frequency. (d) Electric and magnetic dipole moments excited within the resonators for the harmonic with $n = +1$. (e) Forward scattering cross section and the forward-to-backward ratio for the harmonic with $n = +1$ as a function of frequency and steering angle in the xy-plane, as well as their angular spectra in the resonant condition. (f) Electric field of the excited modes within the resonators and the polar radiation pattern of the ensemble for the harmonic with $n = +1$. This figure elaborates on how time modulation can be used to steer the radiation of the quantum emitter dynamically.

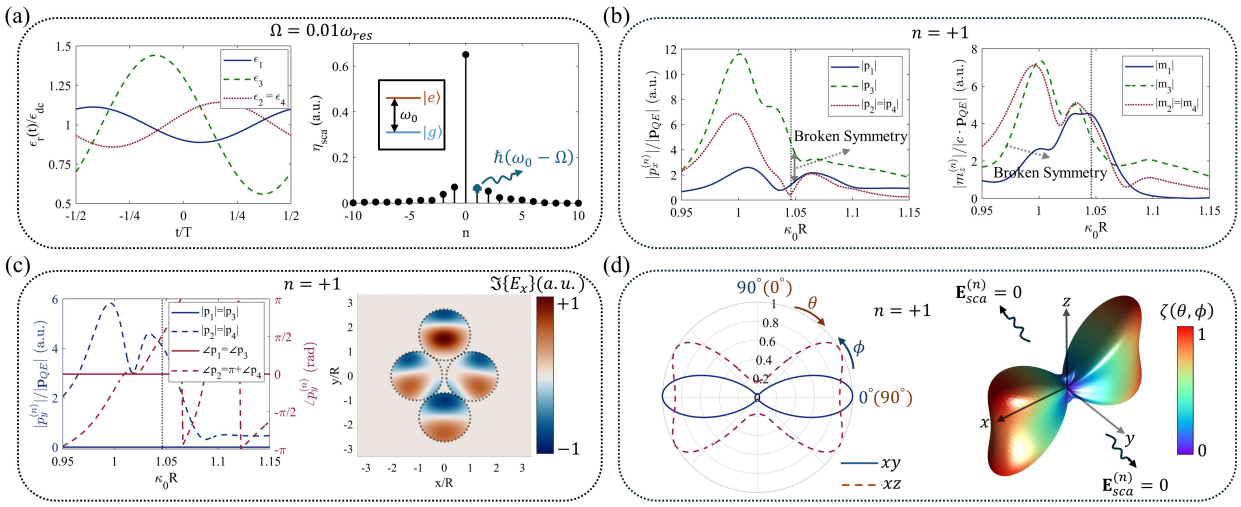


Figure 6: (a) Modulation waveforms within the particles obtained to satisfy transverse Kerker conditions in the xy-plane, and the corresponding scattering efficiency for different harmonics at the resonant frequency. (b) Electric and magnetic dipole moments excited within the resonators for the harmonic with $n = +1$. (c) Electric dipole moments along the y-axis induced in the particles, and the electric field in the polarization direction of the quantum emitter for the harmonic with $n = +1$. (d) Far-field radiation patterns of the ensemble for the harmonic with $n = +1$. This figure elucidates how time modulation can be utilized to satisfy transverse Kerker conditions dynamically.

the xy- and xz-planes, which can be utilized to enable unique on-chip applications in integrated quantum optics and photonics.

4.4 Excitation and application of nonradiating anapole and scattering states as a dynamic switch

Resonators comprised of dielectric materials with high refractive indices support various multipolar resonant modes, and their interactions can bring about numerous promising applications. Among different scattering phenomena, nonradiating anapole states are of particular interest due to the avenues they open toward unique applications in sensing and nonlinear optics [38, 39]. In fact, anapole states can be effectively explained in the context of Cartesian multipole expansion of electromagnetic fields, where the destructive interference of primitive multipolar modes with their toroidal counterparts results in nonradiating modes excited in the system. Anapole states of the resonators are associated with two major attributes: strong enhancement of the electric field and a negligible far-field scattering footprint. Therefore, they can be utilized to enhance the nonlinear effects and sensing capabilities of optical systems.

Time modulation, contingent upon careful selection of the modulation waveforms, can be utilized to dynamically manipulate the interactions between the induced multipoles within different particles, giving rise to a nonradiating anapole state. To give an illustration, the excitation and modulation angular frequencies are considered $\omega_0 = 0.95\omega_r$ and $\Omega = 0.05\omega_r$, respectively. Consequently, the harmonic with $n = -1$ radiates at the resonant frequency of the cavity, which is the main focus of this section. As demonstrated in Figure 7a, the inversion symmetry of the ensemble along the x- and y-axes is preserved by using identical modulation functions for the particles placed on the same axis ($\epsilon_1 = \epsilon_3$ and $\epsilon_2 = \epsilon_4$). This symmetry ensures that the excited modes are solely of even or odd parity, allowing them to destructively interfere and result in the nonradiating anapole state. In order to engineer the system's symmetry and excite the nonradiating state of the resonators, the modulation strength and phase for particles two and four are fixed at $\delta_2 = 0.1$ and $\phi_{m2} = 0$, respectively. Then, the ratio between the modulation strength of the resonators on opposite axes, δ_1/δ_2, as well as their relative phase, $\Delta\phi_m$ ($\phi_1 - \phi_3$), are varied to determine the parameters that result

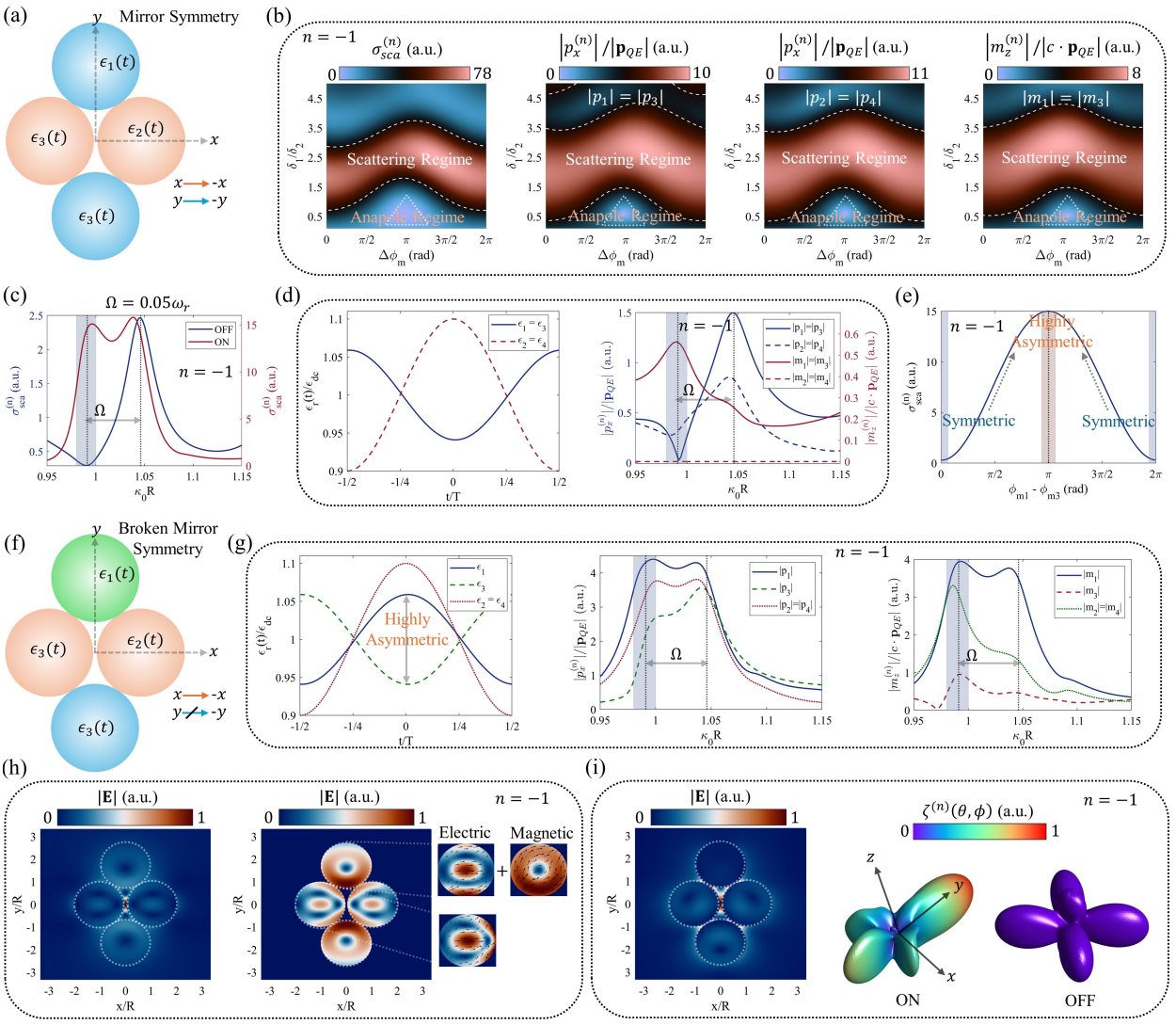


Figure 7: (a) Schematic illustration of the symmetric ensemble of resonators. (b) Scattering cross section and the electric and magnetic dipole moments in the particles for the harmonic with $n = -1$ as a function of the ratio of the modulation strengths of particles one and two and their relative modulation phase. (c) Scattering cross section of the harmonic with $n = -1$ for the OFF and ON states. (d) Modulation functions of the particles optimized to obtain the anapole nonradiating state, and the electric and magnetic dipoles in the particles for the harmonic with $n = -1$ as a function of frequency. (e) Scattering cross section of the harmonic with $n = -1$ as a function of the relative phase difference between the particles on the y-axis. (f) Depiction of the resonators with broken mirror symmetry. (g) Modulation waveforms of the resonators resulting in the scattering state and the corresponding electric and magnetic dipole moments in the particles for the harmonic with $n = -1$. (h) Electric field norms in the anapole state for the harmonic with $n = -1$ and the decomposition into electric and magnetic multipoles. (i) Electric field norm related to the scattering state, and the radiation patterns of ON and OFF states for the harmonic with $n = -1$. This figure elaborates on time modulation as a dynamic switch between the anapole nonradiating and scattering states.

in minimum light scattering. Figure 5b illustrates the scattering cross section and the excited electric and magnetic dipole moments for the time-Floquet mode with $n = -1$ inside the particles with respect to the aforementioned variables. It should be noted that due to the inherent symmetry of the resonators, the amplitudes of the excited electric and magnetic dipole moments are identical for the particles placed on the same axis, and only the particles along the y-axis experience the magnetic modes. As can be seen from Figure 7b, the scattering cross section and the electric and magnetic dipole moments of the particles converge to their minima at $\delta_1/\delta_2 = 0.59$ and $\Delta\phi_m = 3.079\ rad$, indicating the presence of a nonradiating state in the cavity, also referred to as the OFF state. Therefore, the modulation waveforms are chosen accordingly, and the frequency-resolved scattering spectra associated with the nonradiating anapole and scattering states for the harmonic with $n = -1$ are shown in Figure 7c. As depicted, the scattering cross section related to the OFF state tends toward its minimum at $\omega_0 = 0.95\omega_r$, resulting in a nonradiating anapole mode with an angular frequency of $\omega = \omega_r$. It is worth noting that the scattering cross section is slightly greater than zero due to the presence of the magnetic dipole moments, as will be explained further. To elucidate the nature of the excited nonradiating state, the optimal modulation waveforms for different particles and the associated electric and magnetic dipole moments induced within the resonators for the harmonic with $n = -1$ are illustrated in Figure 7d. Clearly, the electric dipole moments for all the particles reach their minimum at the excitation frequency, demonstrating the excitation of the electric dipole anapole state in the resonators. On the contrary, although the induced magnetic dipole moments are minimal, as they originate from the mutual coupling between the particles, they do not exactly reach zero. This results in a minor scattering footprint, which appears as the nonzero scattering cross section. Furthermore, the nonradiating anapole state of the ensemble can be transformed into a strongly scattering state by exploiting the symmetry properties of the system, as explained below.

The broken mirror symmetry in the xy-plane can be utilized to convert a nonradiating anapole state into a radiating state, also referred to as the ON state, by allowing the excitation of modes with both even and odd symmetry [103, 49], whose interactions can enhance the far-field radiation. All the modulation parameters are the same as for the anapole state, except for the relative phase between the particles on the y-axis, which is used to break the mirror symmetry. As demonstrated in Figure 7e, by increasing the relative phase between particles one and three, the system starts to respond to the enhanced asymmetry by radiating to the far-field, reaching its maximum at a relative phase of π, corresponding to a highly asymmetric ensemble. The illustrated scattering cross section of the radiating state in Figure 7c clearly shows a peak at $\omega_0 = 0.95\omega_r$, indicating the system's scattering mode. Moreover, the ensemble with broken symmetry, the modulation waveforms resulting in the scattering state, and the corresponding electric and magnetic dipole moments induced within the resonators are demonstrated in Figures 7f and 7g, respectively. As can be seen, the broken symmetry of the system introduces an additional resonance near $\omega_0 = 0.95\omega_r$, corresponding to the radiating mode of the cavity. Additionally, the excited dipole moments are no longer symmetric, and magnetic modes are observed within the particles on the x-axis. It should be noted that the asymmetry also gives rise to electric dipole moments perpendicular to the polarization direction of the quantum emitter (y-axis), which are not presented here for the sake of brevity (a detailed analysis of the nonradiating anapole and scattering states in the local coordinates of the quantum emitter can be found in the Supplemental Material [88]).

To further elaborate on the nature of the nonradiating state, the electric field norms related to the harmonic with $n = -1$ for the anapole state are illustrated in Figure 7h. As illustrated, the nonradiating electric hotspot formed at the center of the cavity is the signature of the anapole state, which can be exploited to enhance the nonlinear response in optical systems. Additionally, the electric fields inside the particles are decomposed into contributions from the electric and magnetic multipolar modes. Evidently, the excited electric modes within the individual resonators resemble the electric field of an electric dipole anapole state, as anticipated. Finally, the electric field for the harmonic with $n = -1$ in the scattering state, as well as the radiation pattern of the ensemble in both the ON and OFF states, are depicted in Figure 7i. In this scenario, the electric hotspot effectively contributes to the radiation and enhances the spontaneous emission rate of the quantum emitter. Additionally, the radiation patterns clearly show that time modulation can be used as a dynamic switch to convert a nonradiating anapole state into a scattering mode in the ensemble, enabling various integrated quantum physics applications in nonlinear optics and sensing. Finally, while this study primarily focuses on fundamental aspects of quantum emitter control using time-varying Mie resonators, exploring higher modulation frequencies, including parametric amplification effects, presents intriguing opportunities. Such extensions, particularly in the context of time-varying metasurfaces, are part

of our ongoing research.

5 Conclusion

In this study, we introduced Mie resonators composed of materials with time-varying properties to actively enhance and control the spontaneous emission of quantum emitters. Specifically, we utilized a symmetric configuration of four dielectric spherical resonators with time-modulated permittivities to manipulate the spectral and radiation attributes of a quantum dot, modeled by its transition dipole moment. We proposed a comprehensive theoretical platform to capture the light-matter interactions of a quantum emitter with time-varying resonators. Using the time-Floquet method combined with multipole expansion, we derived the dynamic Mie coefficients for a single time-modulated resonator and utilized multipole translation coefficients to describe the full vector coupling between different particles. We elaborated on the design and analysis of a static ensemble of silicon resonators to enhance the spontaneous emission of the quantum dot, utilizing the Purcell factor. Our investigation of the excited modes within the particles demonstrated that these modes can result in remarkably directive radiation. We introduced time modulation to the ensemble as a sinusoidal variation in the permittivity of the particles and used it as an optimal feedback strategy to compensate for frequency detuning in the radiation of the quantum emitter, which arises from its stochastic nature. We illustrated that careful selection of the modulation waveforms and their angular frequency can effectively atone for detuning while preserving the desired radiation pattern. Furthermore, we showed that time modulation can be exploited to break the symmetry of the structure, allowing for the excitation of multipolar resonant modes necessary to achieve the desired radiation characteristics. In this context, we carefully controlled the interactions between the excited modes in the resonators, steering the quantum emitter's radiation toward various angles. We introduced transverse Kerker conditions and demonstrated that time modulation can dynamically satisfy these conditions, thereby rotating the radiation of the quantum emitter toward its polarization direction. Finally, we illustrated that, in the presence of mirror symmetry, a nonradiating anapole state can be excited in the ensemble by suppressing the multipolar modes within each individual resonator. We traced the origins of this nonradiating state to the electric dipole anapole induced in each resonator and demonstrated that it can enhance the electric field while suppressing far-field radiation. By breaking the mirror symmetry through time modulation, we excited multipolar resonant modes with mixed parity, converting the anapole state to a radiating state. We demonstrated that by imposing a phase difference between the resonators, time modulation can serve as a dynamic switch to transition between nonradiating and radiating states, which is crucial for enhanced sensing and nonlinear effects. Our results suggest a comprehensive and practical pathway for dynamic control of the spectral and radiation attributes of quantum emitters, opening up avenues for numerous applications in integrated quantum physics. Our method can be further expanded by studying the impacts of a time-varying cavity on the transition of electrons in multilevel systems, which will be addressed in future works.

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